Jit Malla

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Research Objective

Computer Science graduate and current **Ph.D. Research Scholar in Computer Science and Engineering at IIT Kharagpur** with strong interests in **computer vision**, **machine learning**, **deep learning**, and **scene graph–based visual reasoning**. My research focuses on bridging visual perception and symbolic reasoning through deep learning, graph-based representations, and vision-language models. I am particularly interested in understanding the **robustness, reliability, interpretability, and limitations of multimodal AI systems**, with an emphasis on multimodal learning, visual reasoning, and intelligent systems with real-world impact.

Research Experience

M.Sc. Research Project — IACS Kolkata 2024 – 2026
Scene Graph Based Spatial Reasoning and Visual Grounding in Laboratory Environments using Vision-Language Models

- Developed a pipeline for **scene graph generation** from chemistry laboratory images using vision-language models under two settings: image-only and image with bounding box annotations.
- Conducted a systematic study demonstrating that **bounding box annotations improve object identification and spatial reasoning**, with measurable gains in recall and F1-score.
- Generated structured **scene graphs containing objects and spatial relations** and designed post-processing methods for improved **geometric consistency and relational accuracy**.
- Leveraged **vision-language models and LLMs for spatial reasoning and question answering** over generated scene graphs, incorporating verification mechanisms to improve logical consistency.
- Investigated challenges related to **model reliability, error propagation, and hallucination** in multimodal pipelines, with ongoing research toward **real-time and layered 3D scene graph representations**.

B.Sc. Project — Ramakrishna Mission Residential College, Narendrapur 2023 – 2024
A Wavelet-Based Approach for Authenticating Medical Images and Extracting Patient Information
Guide: Bibek Ranjan Ghosh, Assistant Professor

- Developed a secure medical image authentication framework using **wavelet transforms** for embedding and extracting patient metadata.
- Ensured preservation of **diagnostic image quality** while maintaining data integrity.
- Evaluated robustness against **compression, noise, and common image processing distortions**.

Technical Skills

Programming Languages: Python, C, C++, Java
Machine Learning & Computer Vision: Design and implementation of Convolutional Neural Networks (CNNs), Feature Extraction, Latent Space Visualization using t-SNE, Transformer-based Models, Deep Learning
Graph-Based Vision & Reasoning: Scene Graph Generation, Object–Attribute Relationship Modeling, Spatial Relationship Inference, Graph-based Knowledge Representation
Tools & Frameworks: PyTorch, NumPy, OpenCV, Transformers, YOLO, Hugging Face, Visual Studio Code, Linux/Unix
Documentation & Visualization: LaTeX, Graph and Scene Graph Visualization using PyDot

Relevant Coursework

Computer Science: Data Structures and Algorithms, Object-Oriented Programming (Java), DBMS, Operating Systems, Theory of Computation, Digital Image Processing, Algorithm Analysis, Software Engineering, Computer Networks, Artificial Intelligence & Data Science, Discrete Mathematics, Computer Architecture, Digital Electronics, Numerical Analysis, Machine Learning, Artificial Intelligence

Mathematics: Linear Algebra, Ordinary Differential Equation
Statistics: Descriptive Statistics (I & II), Probability Theory (I & II), Sampling Methods, Statistical Inference

Additional Information

- Qualified **GATE 2026 (Computer Science and Information Technology)**.
- Currently pursuing a **Ph.D. in Computer Science and Engineering at IIT Kharagpur**.
- Completed the **M.Sc. in Computer Science at IACS** with a final CGPA of **8.15/10**.
- Research interests include **computer vision, machine learning, deep learning, graph-based representations, and vision-language reasoning frameworks**, with hands-on experience in scene graph-based spatial reasoning.
- Experienced in working within research-driven and collaborative academic environments, including iterative experimentation, research presentations, and discussion of research ideas.